

ERIC MAY

Software Engineer · Unity & C# · Backend & Cloud · Simulation & Ed-Tech

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Based in Blacksburg, VA · Open to Remote

US citizen, eligible for security clearance · Open to full-time, contract, or contract-to-hire

PROFESSIONAL SUMMARY

Software engineer and technical educator with 10 years of applied CS experience spanning Unity/C# game and simulation systems, full-stack development, curriculum engineering, and developer tooling. Built and shipped production software—including a cloud-hosted leaderboard backend (FastAPI, PostgreSQL, Heroku) and a commercial game on Steam and Google Play—while simultaneously designing and teaching rigorous, project-driven CS coursework at the university level. Proven ability to architect systems, communicate complex technical concepts to diverse audiences, and drive measurable outcomes across engineering and education contexts.

TECHNICAL SKILLS

Languages	C#, Python, C++, C, SQL, TypeScript, JavaScript
Engines / Frameworks	Unity (6, URP, Input System), FastAPI, React, Vite
Data	PostgreSQL, Redis, Jinja2
Cloud / Infra	Heroku, AWS (EC2, RDS, S3), Docker, Linux
Testing / CI	Pytest, GitHub Actions, Sentry
Tooling	Git, REST APIs, automated grading infrastructure
Disciplines	Game Development, Simulation Systems, Full-Stack Dev, Curriculum Engineering, Technical Writing, Developer Advocacy

SELECTED PROJECTS

High Score Server (Full-Stack Cloud Backend) · [Live](#) · [Repo](#)

- FastAPI / Python REST backend supporting persistent leaderboards for Unity games; deployed on **Heroku** with a **PostgreSQL** database and a swappable cache backend (Redis or in-process TTL).
- Unity client integration via a drop-in C# SDK with coroutine-based API calls, typed response models, and a result-wrapper pattern that surfaces errors without exceptions; handles the full auth lifecycle including silent guest login, token storage, and account claiming.
- Architecture handles concurrent score submissions, authenticated writes, ranked reads via SQL window functions, and IP-based rate limiting—designed to support multiple games from a single deployment.
- Production observability via Sentry with sampled performance tracing; integration test suite via pytest against an ephemeral Postgres service, executed on GitHub Actions on every push and PR; deploys to Heroku continuously from main.
- Server-rendered public leaderboard views via Jinja2 plus a React + TypeScript single-page client.

Minesweeper Madness (Shipped Commercial Game) · [Steam Store](#)

- Solo-developed and shipped on Steam and Google Play; implemented classic and novel gameplay modes.
- Wrote a [platform-lifecycle devlog](#) reflecting on Steam-vs-Play maintenance tradeoffs and the engineering case for platform consolidation.

Intro Game Dev Course (Unity Curriculum) · github.com/EricMay256/intro-gamedev-course

- Solo-authored Unity 6 / URP project backbone for a university game development course; 12 progressive lessons covering physics, scripting patterns, tilemaps, audio, animation, procedural generation, debugging tools, shaders, and multiplayer.
- Used by **550+ students** across multiple cohorts; published as a public MIT-licensed template repository so other educators and self-learners can fork it as a starting point.

UBear (Unity Utility Library) · github.com/EricMay256/UBear

- Personal cross-project Unity library—reusable systems, components, and tooling, including the C# SDK client for the High Score Server. Designed as mechanism rather than policy: composable primitives, not opinionated frameworks.
- Published as a versioned UPM package (`com.ericmay.ubear`) installable via Unity Package Manager from the repository's git URL; MIT-licensed, semver-tagged, with separated runtime and test assemblies.

PROFESSIONAL EXPERIENCE

Lecturer — Computer Science · *California State University, Fullerton* 2016 – Present

- Designed and delivered courses across the CS curriculum—including game development, algorithms, systems, and introductory programming—taught approximately 2,700 students across a decade of instruction; consistently improving student pass rates and project quality through iterative course redesign.
- **Rebuilt the introductory Game Development course from the ground up:** replaced theory-only content with a hands-on Unity/C# project track, driving a significant increase in enrollment demand and measurable improvement in student-shipped project quality.
- Managed 3–5 sections per semester (up to ~45 students each); built automated grading infrastructure in Python and C++, with GitHub Classroom integrations, to assess code submissions at scale, cutting grading cycle time and improving feedback consistency.
- Introduced Git-based assignment workflows, normalizing version control habits in students with no prior experience—a transferable skill explicitly requested by industry partners.
- Designed and deployed an AI academic integrity framework: syllabus policy, structured assignments, and evaluation rubrics for responsible AI use—adopted ahead of most peer institutions.

Independent Developer / Contractor · *Self* 2013 – Present

- Contract engagements spanning full application development (ported legacy VB.NET application to C# with SQL database layer) and playable game prototype delivery for external clients.
- Full independent ownership of personal projects—architecture, scope, version control, and post-launch support—across Unity/C# game and tooling work.

Back-End Software Developer · *Eddi* May 2015 – Nov 2015

- Full-time engineering role on a multi-platform marketplace application; contributed across back-end (Go, PostgreSQL), front-end (React, Redux), and cloud infrastructure (AWS: EC2, RDS, S3, Elastic Beanstalk; Docker).
- Integrated Braintree payments and collaborated with stakeholders to translate product requirements into technical implementations across the multi-system solution.
- Administered company cloud accounts and managed environment configuration on AWS.

EDUCATION

B.S., Computer Science · *California State University, Fullerton* · 2015

Admitted with Distinction; Dean's List